

Inter-Process Communication

Inter-process communication, or IPC, is the way separate processes on a system pass data to each other and keep their actions in sync. Because the OS keeps every process in its own protected memory, they can't just read each other's variables, so it provides channels like pipes, message queues, shared memory, semaphores, signals, and sockets to let them talk.

IPC techniques:

There are in total 8 IPC techniques in which 2 are as follows:

1)Pipes:

A pipe is a one-way channel that lets one process hand data to another. Whatever the first process writes into one end comes out the other end for the second process to read, in the same order it went in. The OS holds the data in a small buffer in between, so the two processes don't have to run at exactly the same speed.

Example:When you search a list of files for a certain word, one program produces the list and a second program filters it. The pipe connects them so the list flows straight into the filter, with no need to save anything to a temporary file first.

2)Named pipes:

A named pipe, also called a FIFO (first in, first out), is a pipe that has a name and appears in the file system like a regular file. Because it has a name, any process that knows it can open it, so even completely unrelated programs can use it to send data to each other. It's still one-way, and data comes out in the same order it went in.

Example:A report generator writes finished reports into a named pipe called "print-queue." A separate print manager, started later by someone else, opens that same pipe by name and prints the reports in the order they arrived. The two programs never launched each other and only share the pipe's name.